

## Second Assignment: Dataflow analysis

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# 1 Very Busy Expressions

An expression is defined as a *very busy expression* at a point  $p$  if any path starting from  $p$  uses the expression without redefining any of its operands. To identify them, we proceed backward, starting from the *Exit* basic block of the Control Flow Graph (CFG), applying the transfer function and using the intersection ( $\cap$ ) as the meet operator.

| Parameter                                   | Definition                                                      |
|---------------------------------------------|-----------------------------------------------------------------|
| <b>Domain</b>                               | Sets of expressions                                             |
| <b>Direction</b>                            | Backward: $in[b] = f_b(out[b])$ , $out[b] = \wedge in[succ(b)]$ |
| <b>Transfer Function</b>                    | $in[b] = Gen_b \cup (out[b] \setminus Kill_b)$                  |
| <b>Meet Operation (<math>\wedge</math>)</b> | Intersection ( $\cap$ )                                         |
| <b>Boundary Condition</b>                   | $in[Exit] = \emptyset$                                          |
| <b>Initial Interior Points</b>              | $in[B] = U$ (Universal set)                                     |

Table 1: Very Busy Expressions parameters

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## Algorithm 1 Very Busy Expressions Algorithm

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```

1:  $in[Exit] \leftarrow \emptyset$ 
2: for each basic block  $B \neq Exit$  do
3:    $in[B] \leftarrow U$ 
4: end for

5:  $changes \leftarrow true$ 
6: while  $changes$  do
7:    $changes \leftarrow false$ 
8:   for each basic block  $B \neq Exit$  do
9:      $out[B] \leftarrow \bigcap_{a \in succ(B)} in[a]$ 
10:     $in_{new} \leftarrow Gen_B \cup (out[B] \setminus Kill_B)$ 
11:    if  $in[B] \neq in_{new}$  then
12:       $in[B] \leftarrow in_{new}$ 
13:       $changes \leftarrow true$ 
14:    end if
15:   end for
16: end while

```

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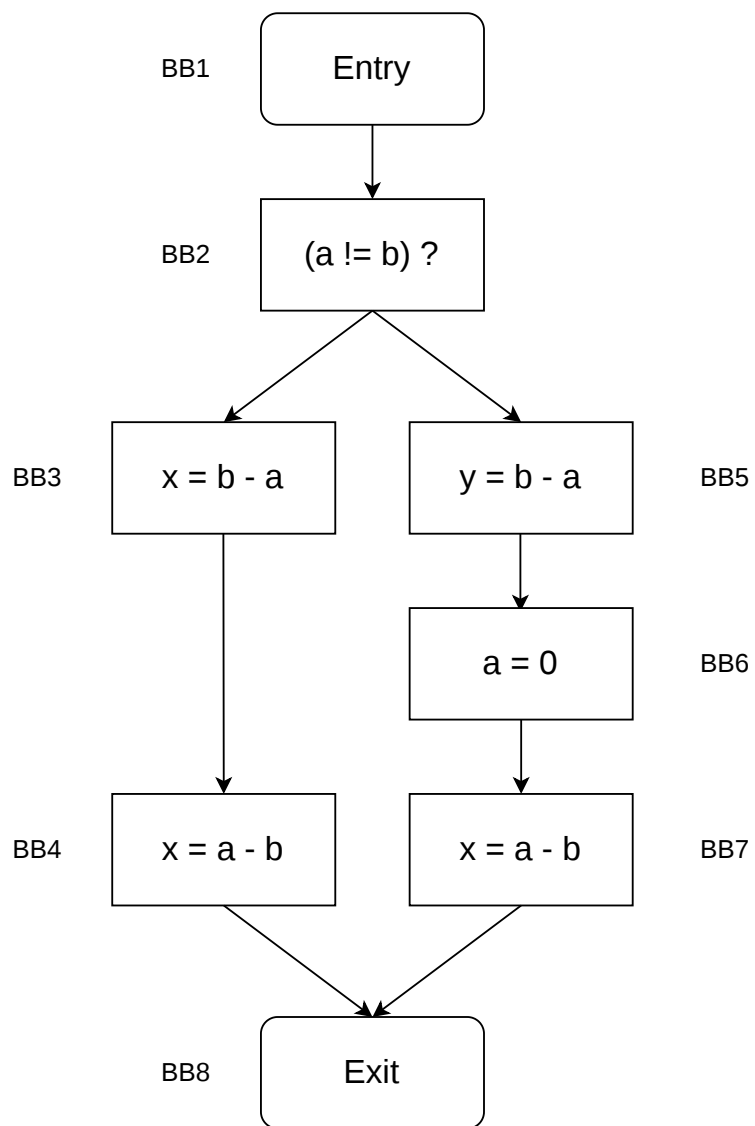


Figure 1: Very busy expression example CFG

| Basic Block        | Iteration 1         |                     | Iteration 2         |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|
|                    | IN[B]               | OUT[B]              | IN[B]               | OUT[B]              |
| <b>BB1 (Entry)</b> |                     | $\{b - a, a! = b\}$ |                     | $\{b - a, a! = b\}$ |
| <b>BB2</b>         | $\{b - a, a! = b\}$ | $\{b - a\}$         | $\{b - a, a! = b\}$ | $\{b - a\}$         |
| <b>BB3</b>         | $\{b - a, a - b\}$  | $\{a - b\}$         | $\{b - a, a - b\}$  | $\{a - b\}$         |
| <b>BB4</b>         | $\{a - b\}$         | $\emptyset$         | $\{a - b\}$         | $\emptyset$         |
| <b>BB5</b>         | $\{b - a\}$         | $\emptyset$         | $\{b - a\}$         | $\emptyset$         |
| <b>BB6</b>         | $\emptyset$         | $\{a - b\}$         | $\emptyset$         | $\{a - b\}$         |
| <b>BB7</b>         | $\{a - b\}$         | $\emptyset$         | $\{a - b\}$         | $\emptyset$         |
| <b>BB8 (Exit)</b>  | $\emptyset$         |                     | $\emptyset$         |                     |

Table 2: Iteration table for example CFG (we iterate on BBs in decreasing order)

## 2 Dominator Analysis

L'analisi dei Dominatori è fondamentale per la costruzione della Static Single Assignment (SSA) form e per l'individuazione dei loop naturali nel Control Flow Graph. Un nodo  $d$  domina un nodo  $n$  se ogni percorso dal nodo di ingresso al nodo  $n$  deve passare per  $d$ . Spiegazione: Forward perché i dominatori di un nodo dipendono da chi lo precede, non da chi lo segue Transfer function: un blocco aggiunge sé stesso all'insieme dei dominatori in ingresso Boundary: l'Entry domina solo sé stesso all'inizio Meet = intersezione: un nodo  $d$  domina  $b$  solo se domina tutti i predecessori di  $b$  Initial =  $U$ : iniziamo pessimisticamente (ogni nodo è dominato da tutti), poi l'intersezione raffina

| Parameter                                   | Dominator Analysis                                             |
|---------------------------------------------|----------------------------------------------------------------|
| <b>Domain</b>                               | Sets of nodes (subsets of all basic blocks)                    |
| <b>Direction</b>                            | Forward: $out[b] = f_b(in[b])$ , $in[b] = \wedge out[pred[b]]$ |
| <b>Transfer Function</b>                    | $out[b] = \{b\} \cup in[b]$                                    |
| <b>Meet Operation (<math>\wedge</math>)</b> | Intersection ( $\cap$ )                                        |
| <b>Boundary Condition</b>                   | $out[Entry] = \{Entry\}$                                       |
| <b>Initial Interior Points</b>              | $out[B] = U$ (Universal set)                                   |

Table 3: Dominator Analysis as a Dataflow problem

Prima aggiungi subito  $B$  (Gen del nodo stesso) Intersechi tutti i predecessori (Meet) Ripeti su tutti i BB finché nessun  $Dom(B)$  cambia (convergenza)

Si parte dal massimo possibile (AllNodes) L'intersezione può solo ridurre gli insiemi, mai aumentarli Gli insiemi sono finiti  $\rightarrow$  si converge in al più  $|AllNodes|$  iterazioni

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**Algorithm 2** Dominator Analysis Algorithm

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```
1:  $OUT[Entry] \leftarrow \{Entry\}$ 
2: for each basic block  $B \neq Entry$  do
3:    $OUT[B] \leftarrow U$ 
4: end for

5:  $changes \leftarrow true$ 
6: while  $changes$  do
7:    $changes \leftarrow false$ 
8:   for each basic block  $B \neq Entry$  do
9:      $IN[B] \leftarrow \bigcap_{p \in pred(B)} OUT[p]$ 
10:     $OUT_{new} \leftarrow \{B\} \cup IN[B]$ 
11:    if  $OUT[B] \neq OUT_{new}$  then
12:       $OUT[B] \leftarrow OUT_{new}$ 
13:       $changes \leftarrow true$ 
14:    end if
15:   end for
16: end while
```

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### 3 Constant Propagation

Constant Propagation is a global dataflow analysis designed to discover variables that are guaranteed to hold a single constant value at specific program points across all possible execution paths. By tracking pairs of  $\langle \text{variable}, \text{constant} \rangle$ , this analysis enables the compiler to substitute variable uses with their actual values and statically fold binary expressions whose operands are known constants, thereby optimizing execution efficiency.

| Parameter                                   | Definition                                                    |
|---------------------------------------------|---------------------------------------------------------------|
| <b>Domain</b>                               | Sets of pairs $\langle \text{variable}, \text{value} \rangle$ |
| <b>Direction</b>                            | Forward                                                       |
|                                             | $in[B] = \wedge(out[pred(B)])$                                |
|                                             | $out[B] = f_B(in(B))$                                         |
| <b>Transfer Function</b>                    | $f_B(x) = Gen_B \cup (x \setminus Kill_B)$                    |
| <b>Meet Operation (<math>\wedge</math>)</b> | Intersection ( $\cap$ )                                       |
| <b>Boundary Condition</b>                   | $in[entry] = \emptyset, out[entry] = \emptyset$               |
| <b>Initial Interior Points</b>              | $in[B] = U, out[B] = \emptyset$                               |

Table 4: Constant Propagation parameters

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**Algorithm 3** Constant Propagation

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```
1:  $in[Entry] \leftarrow \emptyset$ 
2:  $out[Entry] \leftarrow \emptyset$ 

3: for each basic block  $B \neq Entry$  do
4:    $in[B] \leftarrow U$ 
5:    $out[B] \leftarrow \emptyset$ 
6: end for

7:  $changes \leftarrow \text{true}$ 
8: while  $changes$  do
9:    $changes \leftarrow \text{false}$ 
10:  for each basic block  $B \neq Entry$  do
11:    // Meet operation: combine constants from all predecessors
12:     $in[B] \leftarrow \bigcap_{p \in pred(B)} out[p]$ 

13:    // Transfer function: apply constant folding
14:     $out_{new} \leftarrow f_B(in[B])$ 

15:    if  $out[B] \neq out_{new}$  then
16:       $out[B] \leftarrow out_{new}$ 
17:       $changes \leftarrow \text{true}$ 
18:    end if
19:  end for
20: end while
```

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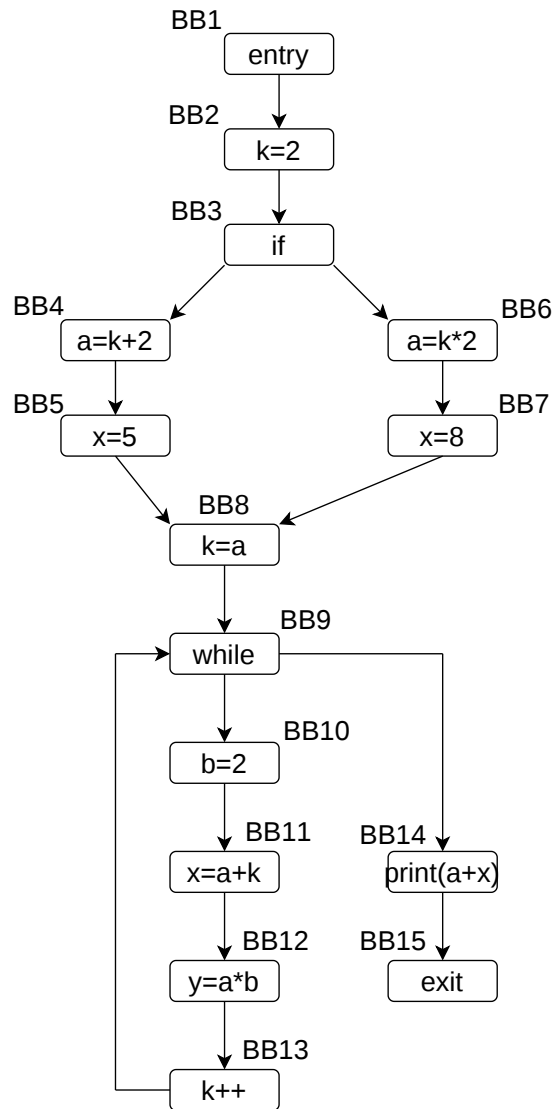


Figure 2: Constant Propagation example CFG

| Basic Block                                  | Iteration 1                                                         |                                                                    | Iteration 2                                  |                                                                    |
|----------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------|
|                                              | IN[B]                                                               | OUT[B]                                                             | IN[B]                                        | OUT[B]                                                             |
| BB1 (Entry)                                  | $\emptyset$                                                         | $\emptyset$                                                        | $\emptyset$                                  | $\emptyset$                                                        |
| BB2                                          | $\emptyset$                                                         | $\langle k, 2 \rangle$                                             | $\emptyset$                                  | $\langle k, 2 \rangle$                                             |
| BB3                                          | $\langle k, 2 \rangle$                                              | $\langle k, 2 \rangle$                                             | $\langle k, 2 \rangle$                       | $\langle k, 2 \rangle$                                             |
| BB4                                          | $\langle k, 2 \rangle$                                              | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle k, 2 \rangle$                       | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       |
| BB5                                          | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle, \langle x, 5 \rangle$ | $\langle k, 2 \rangle, \langle a, 4 \rangle$ | $\langle k, 2 \rangle, \langle a, 4 \rangle, \langle x, 5 \rangle$ |
| BB6                                          | $\langle k, 2 \rangle$                                              | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle k, 2 \rangle$                       | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       |
| BB7                                          | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle, \langle x, 5 \rangle$ | $\langle k, 2 \rangle, \langle a, 4 \rangle$ | $\langle k, 2 \rangle, \langle a, 4 \rangle, \langle x, 8 \rangle$ |
| BB8                                          | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle k, 2 \rangle, \langle a, 4 \rangle$ | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       |
| BB9                                          | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                                             |
| BB10                                         | $\langle k, 4 \rangle, \langle a, 4 \rangle$                        | $\langle k, 4 \rangle, \langle a, 4 \rangle, \langle b, 2 \rangle$ | $\langle a, 4 \rangle$                       | $\langle a, 4 \rangle, \langle b, 2 \rangle$                       |
| BB11                                         | $\langle k, 4 \rangle, \langle a, 4 \rangle, \langle b, 2 \rangle$  | $\langle k, 4 \rangle, \langle a, 4 \rangle,$                      |                                              |                                                                    |
| $\langle b, 2 \rangle, \langle x, 8 \rangle$ | $\langle a, 4 \rangle, \langle b, 2 \rangle$                        | $\langle a, 4 \rangle, \langle b, 2 \rangle$                       |                                              |                                                                    |
| BB12                                         | $\langle k, 4 \rangle, \langle a, 4 \rangle,$                       |                                                                    |                                              |                                                                    |
| $\langle b, 2 \rangle, \langle x, 8 \rangle$ | $\langle k, 4 \rangle, \langle a, 4 \rangle, \langle b, 2 \rangle,$ |                                                                    |                                              |                                                                    |
| $\langle x, 8 \rangle, \langle y, 8 \rangle$ | $\langle a, 4 \rangle, \langle b, 2 \rangle$                        | $\langle a, 4 \rangle, \langle b, 2 \rangle, \langle y, 8 \rangle$ |                                              |                                                                    |
| BB13                                         | $\langle k, 4 \rangle, \langle a, 4 \rangle, \langle b, 2 \rangle,$ |                                                                    |                                              |                                                                    |
| $\langle x, 8 \rangle, \langle y, 8 \rangle$ | $\langle k, 5 \rangle, \langle a, 4 \rangle, \langle b, 2 \rangle,$ |                                                                    |                                              |                                                                    |
| $\langle x, 8 \rangle, \langle y, 8 \rangle$ | $\langle a, 4 \rangle, \langle b, 2 \rangle, \langle y, 8 \rangle$  | $\langle a, 4 \rangle, \langle b, 2 \rangle, \langle y, 8 \rangle$ |                                              |                                                                    |
| BB14                                         | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                                             |
| BB15                                         | $\langle k, 2 \rangle, \langle a, 4 \rangle$                        | $\langle k, 2 \rangle, \langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                       | $\langle a, 4 \rangle$                                             |